SQL PROJECT

README & Project Documentation

Project Name	Sales Data Analysis Using SQL
Tool	MySQL Workbench
Primary Language	SQL
Project Type	Data analysis and query practice

1. Project Overview

This project demonstrates how SQL can be used to explore and analyze structured sales data. It brings together commonly used SQL techniques, from basic data retrieval to more advanced analysis using joins, subqueries, common table expressions (CTEs), and window functions.

2. Objectives

• Retrieve and filter records from database tables.

• Summarize sales and other measures using aggregate functions.

• Combine related information using joins.

• Answer analytical questions using subqueries and CTEs.

• Use ranking and window functions to calculate running totals and compare performance.

• Present query results that support data-driven insights.

3. Tools and Technologies

MySQL — database management system used to create and query the database.
MySQL Workbench — graphical interface used to write, execute, and manage SQL scripts.

4. SQL Topics Covered

Topic	What it is used for
SELECT, WHERE, ORDER BY	Retrieve, filter, and sort records.
GROUP BY and aggregate functions	Summarize data using SUM, AVG, COUNT, MIN, and MAX.
JOINs	Combine records from related tables.
Subqueries	Use the result of one query inside another query.
CTEs	Create readable temporary result sets using WITH.

Topic	What it is used for
Window functions	Perform calculations across related rows without collapsing them.
Ranking functions	Rank customers, products, or employees by performance.
Running totals and growth	Analyze cumulative and month-over-month sales patterns.

5. Suggested Project Files

File / Folder	Description
SQL_Project.sql	Main SQL script containing database setup and analysis queries.
SQL_Project_Backup.sql	Database export containing table structure and data.
Screenshots/	Screenshots of queries, tables, and result sets.
README.pdf	Project overview, tools, topics, and execution instructions.

6. How to Run the Project

1. Open MySQL Workbench and connect to your MySQL server.

2. Open SQL_Project.sql using File → Open SQL Script.

3. If the script creates the database and tables, execute the setup statements first.

4. If using an exported backup, restore it through Server → Data Import, then select the self-contained SQL file and start the import.

5. Select the correct database with USE database_name; before running analysis queries.

6. Execute queries individually or in sections and review the Result Grid and Output panel.

7. Expected Deliverables

The completed project folder should contain the main SQL script, a database backup, this README PDF, and screenshots showing important query results. The exact tables and outputs depend on the dataset and the SQL statements included in the final script.

8. Notes

Before submission, replace or confirm the project name, database name, table names, and query list so they match your actual work. Check that the SQL script runs successfully on your system. Do not include passwords or other private connection details in the project files.